Correlated Failure in AI Agents: A Catastrophe Model from Public Benchmark Data

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Abstract

Insurers are starting to cover losses caused by AI agents, but they have no way to price the risk that many insured agents fail at once. This paper builds a catastrophe model for that accumulation risk from public, item-level benchmark data. On the main coding, terminal and customer-service benchmarks, two models fail the same task about twice as often as independence implies. Across eight capability benchmarks, shared task difficulty explains almost all of this (GSO only in part), and models from the same provider add little beyond it. Prompt-injection attacks are the exception. A two-parameter item response model, conformal recalibration and a person-fit novelty screen forecast how often a new model will fail together with deployed agents, using its headline score and a short audit run. In a forward-in-time test on 82 priced model runs from three benchmarks, including those used to develop the method, the 90% forecast bands cover 89% to 92% of outcomes for pairs and for portfolios of three and five agents. The rule was frozen with code hashes and then tested on a benchmark that played no part in choosing it, where pair bands met the criteria and portfolio bands fell short. In a prototype loss model, the 1-in-200 annual capability loss of a 500-deployment book is 1.48 times what independence implies (1.15 to 1.69 across assumptions) and 1.30 times the vendor-concentration view, with the same average loss. The shared hazard is the task, not the vendor, so diversifying vendors buys less protection than insurers assume.

1. Introduction

AI agents now write code, answer customers and move money for thousands of firms at once, and they run on a handful of foundation models. When one input defeats one agent, it may defeat many. That is accumulation risk, the kind that makes catastrophe insurance different from ordinary insurance. A property insurer can survive many independent house fires but not one hurricane that hits every house on its books.

The insurance market for AI is forming without a way to measure this. From January 2026, standard commercial liability renewals began excluding generative AI, and specialist agent insurers launched (Zylos Research, 2026). AIUC estimates that over 90% of insurers' AI agent exposure sits in silent, unpriced cover (Trout et al., 2026). The same report calls for accumulation models and does not offer one. The industry's working assumption is that the risk concentrates by vendor, so that spreading a book across model providers diversifies it (Bolivar, 2026; Mixides, 2026).

This paper tests that assumption and builds the missing model. It treats a benchmark task as a hazard event, the way a property model treats a storm, and asks which agents fail it together. Two features make this possible with public data. First, several agent benchmarks publish results for every model on every task. Second, those results arrive with each new model release, so a forecast can be checked forward in time.

The paper makes three contributions.

1. **A decomposition of joint failure.** Controlling for task difficulty with item response theory, it separates shared hazard (hard tasks defeat everyone) from vendor concentration (models from one provider fail together beyond difficulty). Across eight capability benchmarks the first dominates. The exception is prompt injection, where attacks are written for a vendor.
2. **A forecast for a model with no history.** From a new model's headline score alone, the method forecasts its joint failure with deployed agents and gives calibrated bands. A person-fit screen refers unfamiliar models to an underwriter instead of pricing them by formula. The rule was frozen with code hashes, registered on OSF, and tested on a benchmark that played no part in choosing it.
3. **Catastrophe-model outputs.** A prototype event-loss model turns joint failures into an exceedance curve, average annual loss and 1-in-200 loss, the figures reinsurers and capital rules use.

2. Related work

Correlated errors and monoculture. Systems built on shared models fail the same people and inputs (Bommasani et al., 2022; Toups et al., 2023), and monoculture can lower welfare even when each system improves (Kleinberg & Raghavan, 2021). Language models agree on their errors more as they grow more capable (Goel et al., 2025; Kim et al., 2025). Recent work finds a co-failure ceiling on combining models (Chen, 2026) and convergence among

SWE-bench agents (Liu et al., 2026). These studies establish the phenomenon. None separates task difficulty from vendor effects for insurance, forecasts it for new models, or turns it into loss.

Item response theory for AI evaluation. Item response models estimate task difficulty and model ability from benchmark results (Maia Polo et al., 2024) and predict task-level agent performance (Ge et al., 2026). Lineage helps predict a new model's scores (Tamura et al., 2025). This paper uses the same tools for a different target: the joint failure of pairs and portfolios, with calibrated uncertainty. Person-fit statistics (Drasgow et al., 1985) become an underwriting screen.

Conformal prediction. Split and jackknife conformal methods give distribution-free coverage (Lei et al., 2018; Barber et al., 2021). Here the calibration errors are the realized forward errors of earlier forecasts, which keeps calibration honest as models change.

Catastrophe and cyber accumulation. Cyber insurance faces the same problem of correlated losses from shared technology (Böhme et al., 2019; Lloyd's, 2023; Lopez & Nkameni, 2026). Proposals for insuring AI set out the stack of products and the need for accumulation models (Trout et al., 2026; Zhu, 2026; Xu et al., 2026). This paper supplies such a model and tests it.

3. Data

All data are public and item-level: for every agent and every task, whether the agent succeeded. Nine benchmarks cover six kinds of agent work and one attack channel (Jimenez et al., 2024; Deng et al., 2025; Chan et al., 2024; Barres et al., 2025; Xu et al., 2024; Debenedetti et al., 2024).

Benchmark	Work	Agents	Tasks	Trials	Role in the paper
SWE-bench Verified, same harness (mini-SWE-agent)	Python bug fixes	39 runs, 33 models, 10 providers; plus 134 leaderboard systems	500	1	Decomposition, loss model, forward test
SWE-bench Multilingual	Bug fixes in nine languages	14 runs, 7 providers	300	1	Replication
Terminal-Bench 2.0	Terminal tasks	61 single-model agents, 25 models, 8 providers	84	about 5	Forward test, trial-level model

Benchmark	Work	Agents	Tasks	Trials	Role in the paper
GSO	Performance optimization	27 models, 6 providers	102	1	Preregistered test
tau2-bench	Customer service	8 agents, 5 providers	278	about 4	Replication
TheAgentCompany	Office work	17 agents, 7 providers	175	1	Replication
SWE-bench Pro (public set), SWE-agent	Harder multi-file software tasks	7 agents, 4 providers	730	1	Replication
MLE-bench	Machine learning engineering (Kaggle competitions)	27 agents, 14 base models, 6 providers	75	seeds	Replication
AgentDojo	Prompt-injection attacks on tool use	22 models, 5 providers	629 cases	1	Security channel

Release dates come from vendor announcements for SWE-bench and Terminal-Bench, and from the date results were added to the repository for GSO, which is when an insurer could first see them. The loss model adds three public inputs: enterprise API usage shares (Menlo Ventures, 2025), cyber loss severity quantiles (Cyentia Institute, 2025), and provider outage histories from public status pages for January 2025 to October 2026.

Two data issues matter. SWE-bench Verified contains flawed tasks and some memorized solutions (OpenAI, 2026), which can create artificial shared failure or success; removing the 26 tasks no system ever solved leaves pair results unchanged but cuts all-fail events from 8.6% to 3.6%, and dropping the 43 tasks none of the 39 runs solved leaves the loss ratio of section 5.5 at 1.50. Leaderboard submissions are chosen by their authors, so the analysis prefers same-harness runs.

4. Method

4.1 Shared hazard versus vendor concentration. Let f be agent a's failure on task t (1 or 0, or the failure share over trials). The joint failure multiple of a pair is the mean of f(a,t) f(b,t) over tasks, divided by the product of their failure rates. A multiple above 1 can come from two sources: some tasks are hard for everyone, or the two agents share something beyond difficulty. An item response model separates them (one-parameter in the main SWE-bench decomposition, with the two-parameter form below as a check and for all forecasts):

$$\text{logit } p(j,t) = a(t) \times [\theta(j) - b(t)]$$

where b is task difficulty, a is how sharply the task separates strong from weak agents, and theta is agent ability. With q the fitted failure probabilities, the residual lift is

$$Lift(a,b) = \Sigma_t f(a,t) f(b,t) / \Sigma_t q(a,t) q(b,t)$$

Task parameters are refitted leaving out the pair's providers, so a provider cannot fit its own tasks. Residual lift near 1 means difficulty explains the joint failure. Intervals come from bootstrap resampling (of tasks on SWE-bench Verified, of models elsewhere), and on AgentDojo a permutation test of provider labels checks the vendor gap.

4.2 Forecasting a model with no history. For a new model, the difficulty model is fitted only on models released earlier and built on a different base model. The new model's ability is set so that its expected score matches its headline score, and 3,000 draws of its task outcomes, conditioned on that total, give a predictive distribution of its joint failure with each deployed agent, or with every agent in a portfolio of three or five from distinct base models. Raw draws are too confident, so conformal prediction sets the band. Each calibration error is scaled by the simulated spread plus a relative floor:

$$z = (y - m) / \sqrt{(s^2 + (c \times m)^2)}$$

where m is the forecast median, s its simulated spread, and c is chosen on the calibration set. Once eight earlier models have been forecast, the calibration errors are their realized forward errors, each model weighted equally, with the usual finite-sample correction. Because c is chosen on the same calibration set, the coverage guarantee is approximate rather than exact.

4.3 Novelty screen and preregistration. A model whose failures do not follow its score breaks the forecast. Before pricing, the new model is run on a random audit of tasks (10% of tasks in version 1; at least 50 and at most half of the tasks in version 2), and the person-fit statistic lz (Drasgow et al., 1985) measures how well its results fit the difficulty model. A model below the 5th percentile of earlier models is referred to an underwriter. A run with no output on more than 5% of tasks is referred for data quality. The full rule, its code hashes and its success criteria were frozen before the GSO test (version 1) and amended once afterwards (version 2), with the selection history disclosed. Both versions, with every hashed file, were registered on OSF on November 5, 2026 (osf.io/t8heg); OSF reports the same sha256 for each file.

4.4 Event-loss model. A hazard event is one benchmark task reaching a random share of an insured book (Beta distributed, mean 10%). Each exposed deployment fails according to its model's result on that task. A share of failures become claims (mean 2%, varying by event), and claims follow a lognormal fitted to cyber loss quantiles, net of a deductible and capped by a limit. The same book is priced under three views with identical failure

rates: independent failures, vendor concentration (deployments on one model fail together, models independent), and the observed shared hazard. A second channel simulates provider outages from published histories, with and without common-cause events. The main assumptions without a public source are varied in a sensitivity test; the deductible and the share of deployments without a fallback are held fixed.

5. Results

5.1 The hazard is the task, not the vendor

On SWE-bench Verified, two distinct models fail the same task about twice as often as independence implies, and 8.6% of tasks defeat every one of 39 runs (95% CI 6.2% to 11.0%), against almost none under independence. Difficulty explains nearly all of it. After controlling for difficulty, same-provider pairs show a residual lift of 1.03 (1.01 to 1.05) against 1.02 across providers; the two-parameter model gives 1.02 and 1.00. Different products built on the same base model add a little more (1.05, CI 1.03 to 1.07, from 119 such pairs among 134 leaderboard systems). The same-provider gap on SWE-bench itself (0.016, CI −0.001 to 0.034) is not significant. The pattern repeats across domains:

Benchmark	Joint failure multiple	Residual lift, same vs cross provider	Reading
SWE-bench Verified	about 2	1.03 vs 1.02	Shared hazard
SWE-bench Multilingual	2.0 to 2.2	not separately tested	Shared hazard; whole book fails on the 99th-percentile task
Terminal-Bench 2.0	1.7 to 2.3	same provider 0.99 (0.95 to 1.03); base model 1.04 (0.99 to 1.11)	Shared hazard
GSO	1.09 to 1.17	0.99 overall	Partly consistent; failure rates near 70% cap the multiple
tau2-bench	2.26 (1.91 to 2.63)	excess +0.09 (−0.04 to +0.27)	Shared hazard
TheAgentCompany	1.07 (1.03 to 1.12)	excess +0.003 (−0.011 to +0.020)	Shared hazard; 41% of tasks defeat all 17 agents

Benchmark	Joint failure multiple	Residual lift, same vs cross provider	Reading
SWE-bench Pro	1.13 (1.08 to 1.21)	excess +0.034 (−0.03 to +0.16)	Shared hazard; 41% of tasks defeat all 7 agents
MLE-bench (any medal)	1.30 (1.16 to 1.46)	excess +0.002 (−0.007 to +0.014)	Shared hazard; 26% of competitions defeat all 27 agents
AgentDojo, prompt injection	2.53 cross, 3.23 same	1.15 vs 0.92, gap 0.23 (0.06 to 0.41)	Vendor excess

The practical consequence is that vendor diversification buys little. At matched skill, a book spread over providers cuts its all-fail rate by only 6% to 13% compared with one concentrated on a single provider. Across models, the shared part of failure is larger for stronger pairs: across 734 SWE-bench pairs, the joint failure multiple rises with the pair's success rate (Spearman 0.97, 0.92 to 0.98). This is cross-sectional and mostly what fixed difficulty implies, since the remaining failures of strong agents sit on the tasks that defeat everyone. Residual dependence rises only modestly with ability (Spearman 0.47) and shows no trend with release date (0.02). Stronger agents fail together less often in absolute terms, but more of what remains is correlated.

5.2 Forecasting new models forward in time

Each model was forecast only from models released before it. Point forecasts are accurate: mean absolute percentage error of 3.1% to 3.7% on SWE-bench and 5.4% to 8.2% on Terminal-Bench, for pairs and portfolios. At one cutoff (October 1, 2025), a constant correlation multiplier fitted on older pairs gave 23% and independence 42%. The two-parameter model beat the one-parameter model (held-out log loss 2% to 4% lower, pair error 10% to 27% lower), while provider-family terms and multi-factor lineage priors changed log loss by 0.2% or less. A benchmark run is needed: forecasting from a predecessor's or the market's results, without the new model's score, gave 17% to 20% pair error against 3% with it.

Raw predictive bands are far too narrow, covering 68% to 79% of outcomes at the 90% level. Recalibration fixes this. On the forward test sets (21 SWE-bench runs from October 2025 to May 2026, 22 Terminal-Bench agents from February to April 2026), the final method's 90% bands cover 88%, 88% and 88% on SWE-bench and 92%, 91% and 88% on Terminal-Bench, for pairs and portfolios of three and five. Their median width is 10% to

11% of the forecast on SWE-bench and 22% to 41% on Terminal-Bench, which has fewer tasks. Two simpler fixes failed: weighting calibration models by date or ability, and adjusting forecasts by the joint failure seen on audit tasks.

The screen referred four of 21 SWE-bench runs, whose bands covered only 52% to 61%. The most extreme, Gemini 3.5 Flash, looked like a model that fails on different tasks. It was not: its run produced no patch on 59 of 500 tasks. On the 441 tasks where it submitted a patch, its person-fit is normal (lz of −1.9) and its joint failure is 95% of forecast rather than 86%. A failure to produce output is an operational channel of its own, which the rule now checks first.

Terminal-Bench's repeated trials show that agents vary on a task more than the two-parameter model implies. Adding an agent-by-task random effect (standard deviation about 1.5 on the logit scale) raises raw 90% coverage from between 56% and 65% to between 71% and 76% and cuts five-agent error from 13.4% to 11.4%, but widens the recalibrated bands by 12% to 19%. This variant is exploratory and not part of the frozen rule.

5.3 A frozen-rule test on a benchmark not used to choose it

The method was frozen on November 5, 2026, with code hashes and success criteria: accepted models' 90% bands should cover 85% to 95% of outcomes, referred models should cover less, and 95% bands should cover at least 90%. GSO, whose pair statistics had been described earlier but which played no part in choosing any forecasting rule, was then forecast forward by the date each result appeared: 13 models from February to September 2026. The newest of them solve 76% to 88% of tasks, when no earlier model solved more than 47%, a jump far outside the training range.

Pair bands passed, covering 90% (95% CI 82% to 96%). Portfolio bands fell short, at 82% and 84% for three and five agents. The 95% portfolio bands also missed (88% for both, against at least 90%), and with no model referred, the comparison of referred and accepted models could not be made. The misses came from Claude Opus 4.8 and Claude Sonnet 5, whose joint failure in five-agent portfolios was 13% to 15% below forecast. Both fit the difficulty model poorly (lz of −5.9 and −4.8 on all tasks), but the version 1 screen only starts after eight earlier models on the same benchmark, and seven had been scored. Three of the four newest models were also overpredicted, by 11% to 16% in five-agent portfolios, and stayed inside their bands only because those bands were three to five times wider.

Pooled over the three forward tests, the frozen rule's bands cover the shares below. The pool holds 82 priced runs: it includes the SWE-bench and Terminal-Bench runs used to develop the method, excludes the 4 referred runs, and includes 30 early runs priced before the screen could act.

Band	Pairs	3 agents	5 agents
50%	51% (47 to 55)	46% (40 to 51)	46% (40 to 52)
80%	82% (79 to 86)	79% (74 to 83)	77% (71 to 81)
90%	92% (89 to 94)	89% (86 to 92)	89% (84 to 92)
95%	96% (94 to 97)	94% (91 to 96)	93% (89 to 96)

A version 2 amendment, frozen the same day, audits at least 50 tasks and lets a new benchmark borrow the screen's threshold from benchmarks with the same trial design until it has eight models of its own. It refers Opus 4.8 and Sonnet 5 and lifts GSO's priced coverage to 95%, 91% and 95%, but because it was designed after seeing GSO, that is not a test. Its test is the monthly scoring of new releases from that date, which runs automatically against the frozen code.

5.4 Security and outage channels

Prompt injection behaves differently. On AgentDojo, case difficulty still explains most shared attack success (82% to 84% of the variance in pair excess over independence), but same-provider pairs show a residual lift of 1.15 against 0.92 across providers, and 1.25 for versions of one base model. The gap survives a permutation test ($p = 0.001$) and a check restricted to cases with identical attack text. Part of it comes from attacks that name the target vendor, which is also how real attackers write. Defenses matter more than vendor choice: a tool filter cuts GPT-4o's attack success from 48% to 7%, and the attacks that still succeed fall mostly on cases the undefended model already failed.

Outages are correlated through shared infrastructure. In the year to October 2026, Anthropic had 20 critical incidents (median 0.7 hours), OpenAI 7 full outages (median 0.5 hours) and Google's Gemini API about 5 listed incidents a year (November 2024 to November 2026). From January 2025 to October 2026, two confirmed events took down two of the top three providers at once (the June 12, 2025 Google Cloud failure and the November 18, 2025 Cloudflare failure), 1.09 a year (90% CI 0.19 to 3.44). Critical Anthropic incidents overlapped full OpenAI outages twice, against 0.17 expected by chance.

5.5 From joint failure to loss

The prototype insures 500 deployments spread over seven current models in proportion to enterprise API usage (Anthropic 32%, OpenAI 25%, Google 20%, the rest over four open-weight models). Claim sizes follow a lognormal fitted to cyber losses for firms under $100 million in revenue (median about $70,000); fourteen publicized AI error cases with a disclosed amount have a median of $47,000.

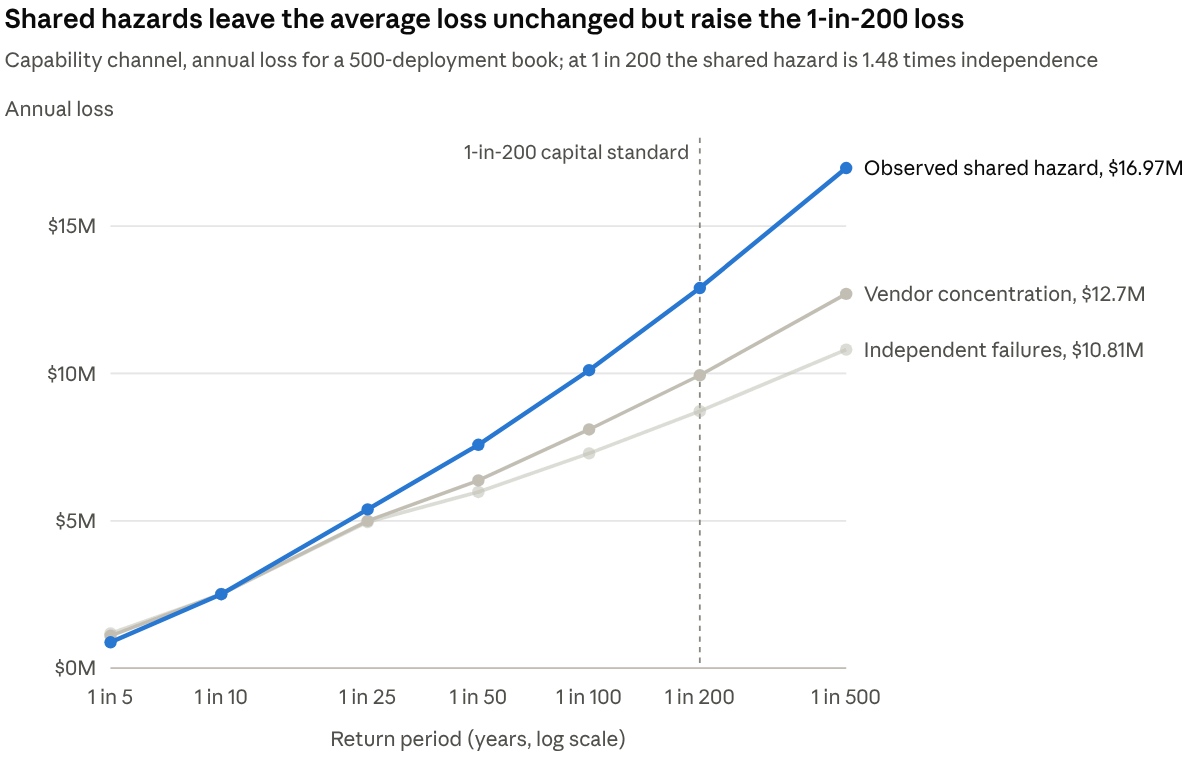


Figure 1 · Prototype loss model, 200,000 simulated years; hazard events drawn from SWE-bench Verified mini-SWE-agent v2.0.0 runs of seven current models weighted by enterprise API share; severity fitted to Cyentia Institute (2025) cyber loss quantiles; event frequency, reach and claim rate assumed

In the capability channel (Figure 1), the three dependence views agree on the average annual loss, about $0.81 million, and part in the tail. The 1-in-200 annual loss is $8.7 million with independent failures, $9.9 million under vendor concentration and $12.9 million with the observed shared hazard, 1.48 times the independent figure and 1.30 times the vendor-concentration figure. The worst event in 200 years takes out 16%, 32% and 48% of the book under the three views. Across one-at-a-time changes to the main unsourced assumptions (event frequency 4 to 36 a year, claim rate 0.5% to 10%, its variation by event, event reach, severity spread, policy limit), the ratio stays between 1.15 and 1.69. With the same assumptions on AgentDojo data, the security channel's ratio is 1.27 for the shared hazard and 1.17 for vendor concentration, against 1.48 and 1.14 for capability: less shared overall, more of it within a vendor.

With an assumed $5,000 an hour of interruption for deployments without a fallback, holding each provider's outage count fixed and moving its share of common-cause events into joint events raises the outage channel's 1-in-200 annual loss from $20.9 million to $22.5 million. Combined, the book's 1-in-200 loss is $22.2 million if every failure is independent and $25.6 million with both observed dependencies, 15% higher; the combined average loss also rises, from $5.9 million to $7.0 million, because shared outages last longer and reach more of the book. The dollar figures depend on assumed frequencies and costs; the ratios between views are the finding.

6. Discussion and limitations

What it means for insurers. Accumulation in AI agents runs through the work, not the vendor. An insurer that limits its exposure per model provider controls the smaller part of the risk. The larger part depends on which kinds of tasks its insureds hand to agents, so the valuable underwriting data is the hazard map of each insured's workload, the same move that made geography the core of property catastrophe models. Three practices follow. Price tail capital on observed joint failure, which here adds about half to the 1-in-200 loss. Require a benchmark run and a short audit before pricing a new model, since forecasts without a score are six times less accurate. Refer models that fail the person-fit screen instead of pricing them by formula. Security is the exception: prompt-injection risk does cluster by vendor, and defenses cut it more than diversification does.

Limitations. Benchmark tasks are not insured losses. A task reaching every agent is a stress scenario, and real customers see different inputs. The loss figures rest on assumed event frequency, claim rate and hourly outage cost; the sensitivity range brackets the ratio, not the dollar amounts. SWE-bench Verified has flawed and memorized tasks, though removing never-solved tasks leaves the pair results intact. Coverage rests on 82 priced runs, and each figure carries an interval of about three to five points. The replications mix setups: TheAgentCompany and MLE-bench combine scaffolds, SWE-bench Pro mixes two run budgets, and every tau2-bench run shares one simulated user. The screen's threshold was chosen among four variants and amended once, which is why the record that counts is the prospective one. GSO's newest models jumped far beyond the training range, and the forecasts there were least reliable. The security results use models a year or more old, and outage counts reflect how providers report incidents.

Next steps. An insurer data partner would replace the assumed parameters with exposure schedules and claims. Further domains with public item-level data (OSWorld and HAL, whose traces are large archives) and a current run of AgentDojo would test the

security exception. The monthly scoring of new releases will show whether the frozen rule holds out of sample.

7. Conclusion

Public benchmark data can price the correlation that AI insurance needs. Joint failure among AI agents is mostly a shared hazard set by the task, it is a larger share of the failures of stronger agents, and it can be forecast for a new model from its headline score with calibrated bands under a rule frozen in advance. In the capability channel, ignoring it understates the 1-in-200 loss of a diversified book by about a third, and the vendor-concentration view understates it by about a quarter, while the average loss is unchanged, which is exactly the error an average-loss view cannot see.

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Appendix: data, code and preregistration

All inputs are public. The code package contains the scripts, result files and the preregistration folder (version 1 rule, version 2 amendment, code hashes and freeze times), registered on OSF on November 5, 2026 (DOI 10.17605/OSF.IO/T8HEG). Benchmark data come from the repositories and dataset linked in section 3; outage histories from the providers' public status pages; loss inputs from the sources cited in section 5.5. Monthly prospective scores of new releases are logged against the frozen code. Target outlets: arXiv (cs.LG, cross-listed to stat.AP, q-fin.RM and cs.CY) and the Geneva Papers on Risk and Insurance special issue on digital and AI transformation in insurance (due March 31, 2027).